

724. Find Pivot Index

Easy Topics Companies Hint

Given an array of integers `nums`, calculate the **pivot index** of this array.

The **pivot index** is the index where the sum of all the numbers **strictly** to the left of the index is equal to the sum of all the numbers **strictly** to the index's right.

If the index is on the left edge of the array, then the left sum is `0` because there are no elements to the left. This also applies to the right edge of the array.

Return the **leftmost pivot index**. If no such index exists, return `-1`.

Example 1:

Input: `nums = [1,7,3,6,5,6]`
Output: `3`
Explanation:
The pivot index is 3.
Left sum = `nums[0] + nums[1] + nums[2] = 1 + 7 + 3 = 11`
Right sum = `nums[4] + nums[5] = 5 + 6 = 11`

Example 2:

Input: `nums = [1,2,3]`
Output: `-1`
Explanation:
There is no index that satisfies the conditions in the problem statement.

Example 3:

Input: `nums = [2,1,-1]`
Output: `0`
Explanation:
The pivot index is 0.
Left sum = `0` (no elements to the left of index 0)
Right sum = `nums[1] + nums[2] = 1 + -1 = 0`

Constraints:

- $1 \leq \text{nums.length} \leq 10^5$
- $-1000 \leq \text{nums}[i] \leq 1000$

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Code

Java Auto

```
1 class Solution {
2     public int pivotIndex(int[] nums) {
3
4         int total = 0;
5
6         // find total sum
7         for (int num : nums) {
8             total += num;
9         }
10
11        int leftSum = 0;
12
13        for (int i = 0; i < nums.length; i++) {
14
15            // Right sum = total - left sum - current number
16            int rightSum = total - leftSum - nums[i];
17
18            if (leftSum == rightSum) {
19                return i;
20            }
21
22            // Add current number to left sum
23            leftSum += nums[i];
24        }
25
26        return -1;
27    }
28 }
29
```

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Testcase Test Result

Accepted Runtime: 0 ms

Case 1 Case 2 Case 3

Input

nums =
[1, 7, 3, 6, 5, 6]

Output

3